

# **Exhibit CC-09**

# 16Gb DDR5 SDRAM Addendum

## MT60B4G4, MT60B2G8, MT60B1G16 Die Revision A

### Features

This document describes the product specifications that are unique to Micron 16Gb DDR5 Die Revision A device. For general Micron DDR5 SDRAM specifications, see the Micron DDR5 SDRAM Core Product Data Sheet. Content in this 16Gb Die Revision A DDR5 SDRAM data sheet addendum supersedes content defined in the core data sheet.

- $V_{DD} = V_{DDQ} = 1.1V$  (NOM)
- $V_{PP} = 1.8V$  (NOM)
- On-die, internal, adjustable  $V_{REF}$  generation for DQ, CA, CS
- 1.1V pseudo open-drain I/O
- TC maximum up to 95°C
  - 32ms, 8192-cycle refresh up to 85°C
  - 16ms, 8192-cycle refresh at >85°C to 95°C
- 32 internal banks (x4, x8): 8 groups of 4 banks each
- 16 internal banks (x16): 4 groups of 4 banks each
- 16n-bit prefetch architecture
- 1 cycle/2 cycle command structure
- 2N mode
- All bank and same bank refresh
- Multi-purpose command (MPC)
- CS/CA training mode
- On-die ECC (bounded fault)
- ECC transparency and error scrub
- Decision feedback equalization (DFE)

- Loopback mode
- Command-based non-target (NT) nominal, DQ/DQS park, and dynamic WR on-die termination (ODT)
- sPPR and hPPR capability
- Per-DRAM addressability
- JEDEC JESD-79.5 compliant

### Options<sup>1</sup>

- **Configuration**
  - 4 Gig x 4
  - 2 Gig x 8
  - 1 Gig x 16
- **FBGA SDP Packages** (Pb-free)
  - x4, x8 82-ball (9mm x 11mm)
  - x16 102-ball (9mm x 14mm)
- **Timing** – cycle time
  - 0.416ns @ CL = 40
- **Operating temperature**
  - Commercial (0°C <  $T_C$  < 95°C)
  - Industrial (–40°C <  $T_C$  < 95°C)
  - Automotive (–40°C <  $T_C$  < 105°C)
- Die Revision

### Marking

4G4  
2G8  
1G16  
  
HB  
HC  
  
-48B  
  
None  
IT  
AT  
:A

Notes: 1. Not all options listed can be combined to define an offered product. Use the part catalog search on [micron.com](http://micron.com) for available offerings.

**Table 1: Key Timing Parameters**

| Speed Grade1 | Speed Bin | Data Rate (MT/s) | Target CL-nRCD-nRP | $t_{AA}$ (ns) | $t_{RCD}$ (ns) | $t_{RP}$ (ns) |
|--------------|-----------|------------------|--------------------|---------------|----------------|---------------|
| -48B         | 4800B     | 4800             | 40-39-39           | 16.000        | 16.000         | 16.000        |

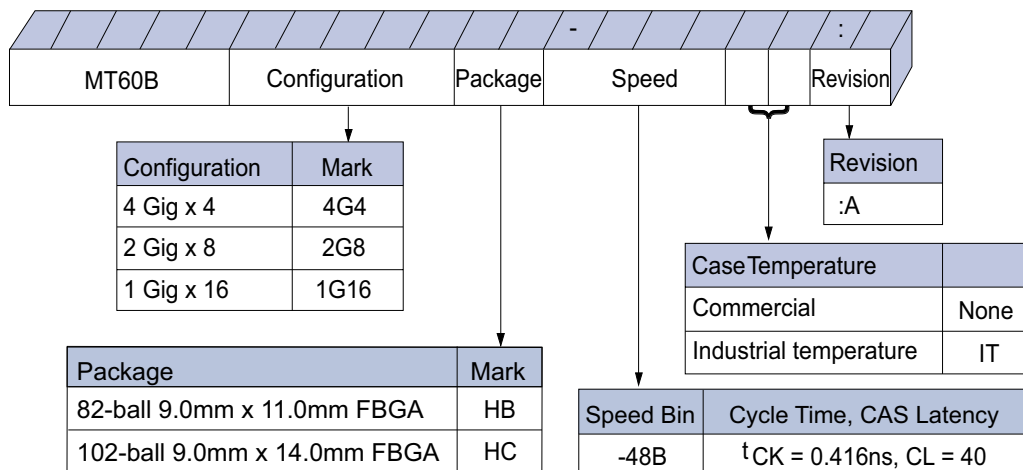
Notes: 1. Refer to the Speed Bin Tables for additional details.

**Table 2: 16Gb Addressing**

| Configuration                 |                                                                      | 4Gb x4       | 2Gb x8       | 1Gb x16      |
|-------------------------------|----------------------------------------------------------------------|--------------|--------------|--------------|
| Bank address                  | Number of bank groups/number of banks per bank group/number of banks | 8 / 4 / 32   | 8 / 4 / 32   | 4 / 4 / 16   |
|                               | Bank group address                                                   | BG0-BG2      | BG0-BG2      | BG0-BG1      |
|                               | Bank address in a bank group                                         | BA0-BA1      | BA0-BA1      | BA0-BA1      |
| Row address                   |                                                                      | R0-R15       | R0-R15       | R0-R15       |
| Column address                |                                                                      | C0-C10       | C0-C9        | C0-C9        |
| Page size                     |                                                                      | 1KB          | 1KB          | 2KB          |
| Chip IDs/maximum stack height |                                                                      | CID0-3 / 16H | CID0-3 / 16H | CID0-3 / 16H |

**Figure 1: Order Part Number Example**

Example Part Number: MT60B2G8HB-48B:A



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## General Notes and Functional Block Diagrams

### General Notes

- The functionality and the timing specifications discussed in this data sheet are for the DLL enable mode of operation (normal operation), unless specifically stated otherwise.
- Throughout the data sheet, the various figures and text refer to DQs as "DQ." The DQ term is to be interpreted as any and all DQ collectively, unless specifically stated otherwise.
- The terms "\_t" and "\_c" are used to represent the true and complement of a differential signal pair. These terms replace the previously used notation of "#" and/or over-bar characters. For example, differential data strobe pair DQS, DQS# is now referred to as DQS\_t, DQS\_c.
- The term "\_n" is used to represent a signal that is active LOW and replaces the previously used "#" and/or overbar characters. For example: CS# is now referred to as CS\_n.
- The terms "DQS" and "CK" found throughout the data sheet are to be interpreted as DQS\_t, DQS\_c and CK\_t, CK\_c respectively, unless specifically stated otherwise.
- Complete functionality may be described throughout the entire document; any page or diagram may have been simplified to convey a topic and may not be inclusive of all requirements.
- Any specific requirement takes precedence over a general statement.
- Any functionality not specifically stated here within is considered undefined, illegal, and not supported, and can result in unknown operation.
- Addressing is denoted as BG[n] for bank group, BA[n] for bank address, and A[n] for row/col address.
- A NOP is considered a valid command for very specific states such as power-down exit, self-refresh exit, and reset. The NOP must satisfy any associated command timings with respect to the preceding valid command.
- Not all features described within this document may be available on the Rev. A (first) version.
- Not all specifications listed are finalized industry standards; best conservative estimates have been provided when an industry standard has not been finalized.
- Although it is implied throughout the specification, the DRAM must be used after reaching a stable power-on level, which is achieved by following the proper voltage ramp and power-up initialization sequence procedures as outline in this specification.
- Not all features designated in the data sheet may be supported by earlier die revisions due to late definition by JEDEC.

### Definitions of the Device-Pin Signal Level

- HIGH: A device pin is driving the logic 1 state.
- LOW: A device pin is driving the logic 0 state.
- High-Z or (HI-Z/Hi-Z): A device pin is tri-state
- ODT: A device pin terminates with the ODT settings, which could be terminating or tri-state depending on the mode register settings.

### Definitions of the Bus Signal Level

- HIGH: One device on the bus is HIGH, and all other devices on the bus are either ODT or High-Z. The voltage level on the bus is nominally  $V_{DDQ}$ .
- LOW: One device on the bus is LOW, and all other devices on the bus are either ODT or High-Z. The voltage level on the bus is nominally  $V_{OL(DC)}$  if ODT was enabled, or  $V_{SSQ}$  if High-Z.
- High-Z or (HI-Z/Hi-Z): All devices on the bus are High-Z. The voltage level on the bus is undefined as the bus is floating.

- ODT: At least one device on the bus is ODT, and all others are High-Z. The voltage level on the bus is nominally  $V_{DDQ}$ .
- The specification requires 8,192 refresh commands within 32ms between 0°C and 85°C. This allows for a  $t_{REFI}$  of 3.9μs in normal refresh mode. The specification also requires 8,192 refresh commands within 16ms between 85°C and 95°C. This allows for a  $t_{REFI}$  of 1.95μs in normal refresh mode.

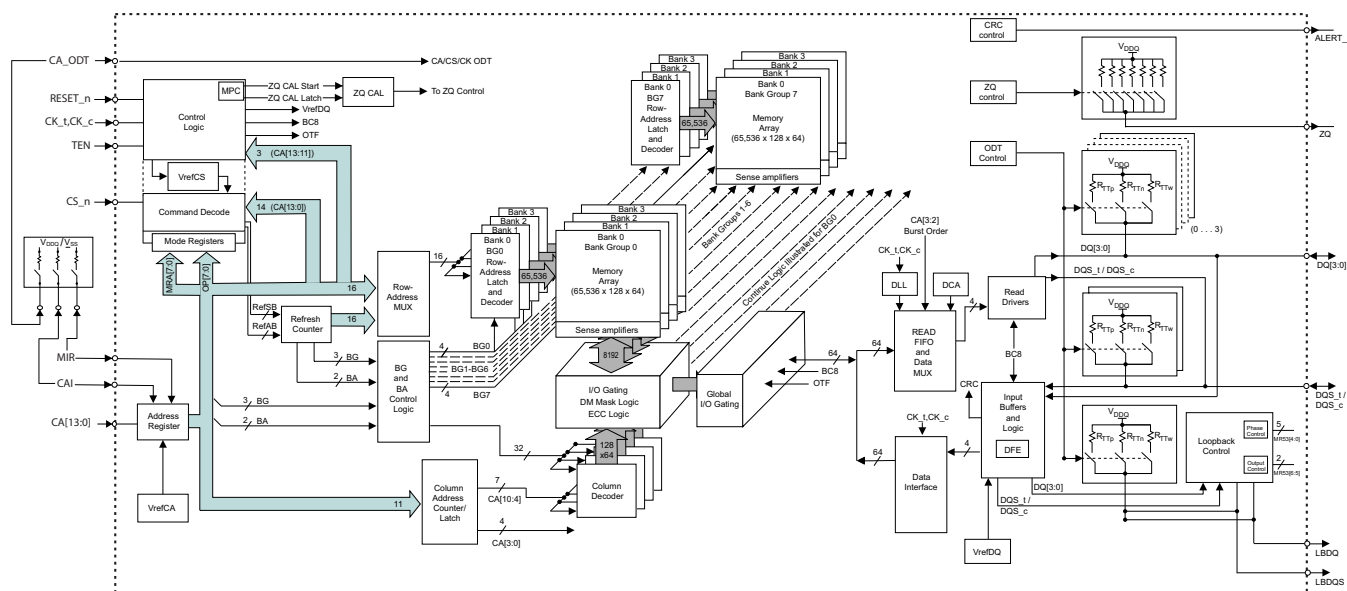
## Industrial Temperature

An industrial temperature (IT) device option requires that the case temperature not exceed below  $-40^{\circ}\text{C}$  or above  $95^{\circ}\text{C}$ . JEDEC specifications require the refresh rate to double when  $T_C$  exceeds  $85^{\circ}\text{C}$ ; this also requires use of the high-temperature self-refresh option. Additionally, ODT resistance and the input/output impedance must be derated when operating outside of the commercial temperature range, when  $T_C$  is between  $-40^{\circ}\text{C}$  and  $0^{\circ}\text{C}$ .

## Automotive Temperature

The automotive temperature (AT) device option requires that the case temperature not exceed below  $-40^{\circ}\text{C}$  or above  $105^{\circ}\text{C}$ . The specifications require the refresh rate to 2X when  $T_C$  exceeds  $85^{\circ}\text{C}$ ; 4X when  $T_C$  exceeds  $95^{\circ}\text{C}$ . Additionally, ODT resistance and the input/output impedance must be derated when operating temperature  $T_C < 0^{\circ}\text{C}$ .

### Figure 2: 4 Gig x4 Functional Block Diagram



[illegible]

## DDR5 Function Matrix

DDR5 SDRAM has several features supported by configuration width, by density, by speed and by device die Rev. The following table is the summary of the features supported by 16Gb Die Revision A by configuration width. The functional matrix will be defined in each device-specific data sheet; therefore, device, speed and density options will vary by device data sheet.

**Table 3: DDR5 Function Matrix - 16Gb Die Rev. A (by configuration width). V: Supported, Blank: Not Supported**

| Function                                                      | x4 | x8 | x16 | MR Default State  | Notes |
|---------------------------------------------------------------|----|----|-----|-------------------|-------|
| BC8 OTF                                                       | V  | V  | V   |                   |       |
| BL32 (JEDEC optional)                                         |    |    |     |                   |       |
| BL32 OTF (JEDEC optional)                                     |    |    |     |                   |       |
| TDQS                                                          |    | V  |     |                   |       |
| Data Mask (DM)                                                |    | V  | V   |                   |       |
| Data Output Disable                                           | V  | V  | V   |                   |       |
| Connectivity Test Mode (CT)                                   | V  | V  | V   |                   |       |
| CA/CS/CK ODT                                                  | V  | V  | V   |                   |       |
| 2N Mode                                                       | V  | V  | V   |                   | 5     |
| Per DRAM Addressability (Enum)                                | V  | V  | V   |                   |       |
| Mode Register Read (MRR)                                      | V  | V  | V   |                   | 3     |
| Mode Register Write (MRW)                                     | V  | V  | V   |                   |       |
| Multi-Purpose Command (MPC)                                   | V  | V  | V   |                   |       |
| ZQ calibration                                                | V  | V  | V   |                   |       |
| CA Vref Training                                              | V  | V  | V   |                   | 1     |
| CS Vref Training                                              | V  | V  | V   |                   | 2     |
| DQ Vref Training                                              | V  | V  | V   |                   |       |
| CS Training Mode (CSTM)                                       | V  | V  | V   |                   |       |
| CA Training Mode (CATM)                                       | V  | V  | V   |                   |       |
| Write Leveling Training                                       | V  | V  | V   |                   |       |
| WICA 1/2 step, a feature add for Internal WL (JEDEC Optional) | V  | V  | V   |                   |       |
| Read Training Pattern Mode (LFSR)                             | V  | V  | V   |                   |       |
| Write Pattern Command                                         | V  | V  | V   |                   |       |
| DQS Interval Oscillator                                       | V  | V  | V   |                   |       |
| Duty Cycle Adjuster (DCA)                                     | V  | V  | V   | MR42:OP[1:0] = 01 | 10    |
| Loopback Mode                                                 | V  | V  | V   |                   |       |
| Decision Feedback Equalization (DFE)                          | V  | V  | V   |                   |       |
| Maximum Power Saving Mode (MPSM)                              | V  | V  | V   |                   | 6     |
| Package Output Driver Test mode (PODTM)                       | V  | V  | V   |                   |       |
| PASR (JEDEC Optional)                                         |    |    |     | MR19:OP[7] = 0    | 11    |
| WRITE CRC                                                     | V  | V  | V   |                   |       |

**Table 3: DDR5 Function Matrix - 16Gb Die Rev. A (by configuration width). V: Supported, Blank: Not Supported (Continued)**

| Function                                                 | x4 | x8 | x16 | MR Default State    | Notes |
|----------------------------------------------------------|----|----|-----|---------------------|-------|
| READ CRC                                                 | V  | V  | V   |                     | 4     |
| Programmable Preamble                                    | V  | V  | V   |                     |       |
| Programmable Postamble                                   | V  | V  | V   |                     |       |
| sPPR                                                     | V  | V  | V   |                     |       |
| hPPR                                                     | V  | V  | V   |                     |       |
| MBIST/mPPR (JEDEC optional)                              |    |    |     | MR23:OP[4:3] = 00   | 12    |
| PPR using DQ[3:0] only                                   | V  | V  | V   |                     |       |
| sPPR undo/lock (JEDEC optional)                          | V  | V  | V   | MR23:OP[2] = 1      | 13    |
| On-Die-ECC                                               | V  | V  | V   |                     |       |
| ECC Transparency and Error Scrub                         | V  | V  | V   |                     |       |
| H-Matrix Revision supporting bounded fault self-aliasing | V  | V  | V   |                     |       |
| Same Bank Refresh                                        | V  | V  | V   |                     | 7     |
| Same Bank Precharge                                      | V  | V  | V   |                     |       |
| Refresh Management (RFM)                                 | V  | V  | V   | MR58:OP[0] = 0      | 9     |
|                                                          |    |    |     | MR58:OP[7:5] = 110  | 14    |
|                                                          |    |    |     | MR58:OP[4:1] = 1010 |       |
|                                                          |    |    |     | MR59:OP[7:6] = 00   |       |
| Adaptive RFM (JEDEC Optional)                            |    |    |     |                     |       |
| Directed RFM (JEDEC Optional)                            |    |    |     |                     |       |
| Fine Granularity Refresh (FGR)                           | V  | V  | V   |                     |       |
| Refresh Interval Rate (RIR) (JEDEC Optional)             |    |    |     | MR4:OP[3] = 0       | 15    |
| Wide Temperature Range (JEDEC Optional)                  |    |    |     | MR4:OP[5] = 0       | 16    |
| Test Mode MR (MR9)                                       |    |    |     |                     | 8     |
| ECS Writeback Suppression (JEDEC Optional)               | V  | V  | V   |                     |       |
| x4 RMW Suppression (JEDEC Optional)                      | V  |    |     |                     |       |

- Notes: 1. CA Vref Training was added to support internally generated CA Vref.  
2. CS Vref Training was added to support internally generated CS Vref.  
3. Mode Register Read (MRR) is similar to DDR4's MPR.  
4. Read CRC as well as Write CRC are supported on DDR5.  
5. 2N Mode replaced what was called Gear Down Mode on DDR4.  
6. MPSM has three states defined: idle, power down and deep power down.  
7. Same Bank Refresh requires FGR be enabled.  
8. Test Mode (TM) is a vendor-specific mode register; not used by Micron.  
9. RFM not required.  
10. Device supports DCA for single/two-phase internal clock(s).  
11. PASR not supported.  
12. MBIST/mPPR not supported.  
13. sPPR lock/undo supported.  
14. RAAMMT, RAAIMT, and RAA counter decrement are only applicable if the RFM requirement bit is set to 1 (MR58:OP[0]=1).



- 15. RIR indicator not implemented.
- 16. Wide temperature range not supported.

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## **DDR5 Package Pinout and Assignments**

### **Rows**

The x4/x8 device has 13 electrical rows of balls. The x16 device has 17 electrical rows of balls. Electrical is defined as rows that contain signal ball or power/ground balls. Additional rows of inactive balls may be available for mechanical support.

### **Ball Pitch**

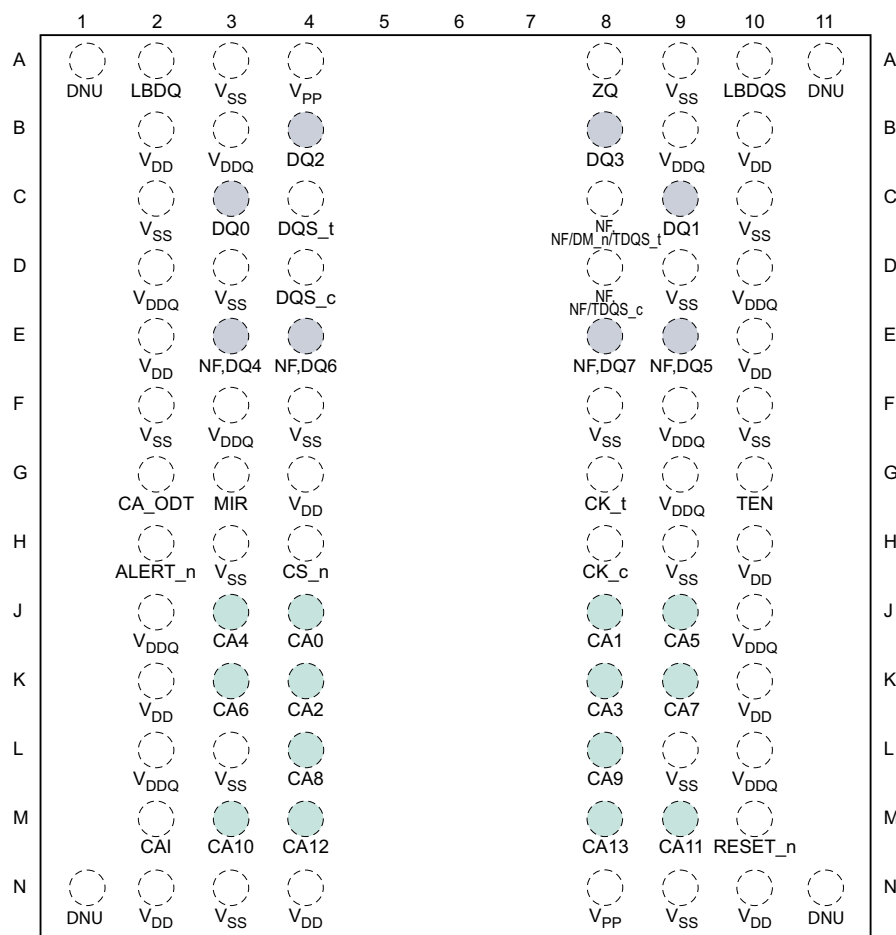
The device uses a ball pitch of 0.8mm x 0.8mm.

### **Columns**

The number of depopulated columns is 3.

The device has six electrical columns of balls in two sets of three columns. Between the electrical columns are three columns where no balls are populated. Electrical is defined as columns that contain signal ball or power/ground balls. Additional columns of inactive balls may be available for mechanical support.

**Figure 5: x4/x8 Ballout Using MO-210-AN – 82-Ball**



- Notes:
1. Additional columns and rows of inactive balls in MO-210-AN terminal pattern (x4/x8) with support balls are for mechanical support only and should not be tied electrically high or low.
  2. Some of the additional support balls can be selectively populated at the suppliers' discretion.
  3. DQ4-DQ7 higher-order DQ pins are connected but not used in the x4 configuration.
  4. DM, TDQS\_t and TDQS\_c are not valid for the x4 configuration.
  5. A comma "," separates the configuration. A slash "/" defines a mode register-selectable function, command/address function, density or package dependence.

**Figure 6: x16 Ballout Using MO-210-AT –102 Ball**

|   | 1                  | 2                | 3                 | 4 | 5 | 6 | 7                | 8                | 9                  |   |
|---|--------------------|------------------|-------------------|---|---|---|------------------|------------------|--------------------|---|
| A | LBDQ               | V <sub>SS</sub>  | V <sub>PP</sub>   |   |   |   | ZQ               | V <sub>SS</sub>  | LBDQS              | A |
| B | V <sub>DD</sub>    | V <sub>DDQ</sub> | DQU2              |   |   |   | DQU3             | V <sub>DDQ</sub> | V <sub>DD</sub>    | B |
| C | V <sub>SS</sub>    | DQU0             | DQSU <sub>t</sub> |   |   |   | DMU <sub>n</sub> | DQU1             | V <sub>SS</sub>    | C |
| D | V <sub>DDQ</sub>   | V <sub>SS</sub>  | DQSU <sub>c</sub> |   |   |   | RFU              | V <sub>SS</sub>  | V <sub>DDQ</sub>   | D |
| E | V <sub>DD</sub>    | DQU4             | DQU6              |   |   |   | DQU7             | DQU5             | V <sub>DD</sub>    | E |
| F | V <sub>DD</sub>    | V <sub>DDQ</sub> | DQL2              |   |   |   | DQL3             | V <sub>DDQ</sub> | V <sub>DD</sub>    | F |
| G | V <sub>SS</sub>    | DQL0             | DQSL <sub>t</sub> |   |   |   | DML <sub>n</sub> | DQL1             | V <sub>SS</sub>    | G |
| H | V <sub>DDQ</sub>   | V <sub>SS</sub>  | DQSL <sub>c</sub> |   |   |   | RFU              | V <sub>SS</sub>  | V <sub>DDQ</sub>   | H |
| J | V <sub>DD</sub>    | DQL4             | DQL6              |   |   |   | DQL7             | DQL5             | V <sub>DD</sub>    | J |
| K | V <sub>SS</sub>    | V <sub>DDQ</sub> | V <sub>SS</sub>   |   |   |   | V <sub>SS</sub>  | V <sub>DDQ</sub> | V <sub>SS</sub>    | K |
| L | CA_ODT             | MIR              | V <sub>DD</sub>   |   |   |   | CK <sub>t</sub>  | V <sub>DDQ</sub> | TEN                | L |
| M | ALERT <sub>n</sub> | V <sub>SS</sub>  | CS <sub>n</sub>   |   |   |   | CK <sub>c</sub>  | V <sub>SS</sub>  | V <sub>DD</sub>    | M |
| N | V <sub>DDQ</sub>   | CA4              | CA0               |   |   |   | CA1              | CA5              | V <sub>DDQ</sub>   | N |
| P | V <sub>DD</sub>    | CA6              | CA2               |   |   |   | CA3              | CA7              | V <sub>DD</sub>    | P |
| R | V <sub>DDQ</sub>   | V <sub>SS</sub>  | CA8               |   |   |   | CA9              | V <sub>SS</sub>  | V <sub>DDQ</sub>   | R |
| T | CAI                | CA10             | CA12              |   |   |   | CA13             | CA11             | RESET <sub>n</sub> | T |
| U | V <sub>DD</sub>    | V <sub>SS</sub>  | V <sub>DD</sub>   |   |   |   | V <sub>PP</sub>  | V <sub>SS</sub>  | V <sub>DD</sub>    | U |

- Notes: 1. Additional columns and rows of inactive balls in MO-210-AU terminal pattern (x16) with support balls are for mechanical support only and should not be tied electrically high or low.
2. Some of the additional support balls can be selectively populated at the suppliers' discretion.

**Table 4: Pinout Description**

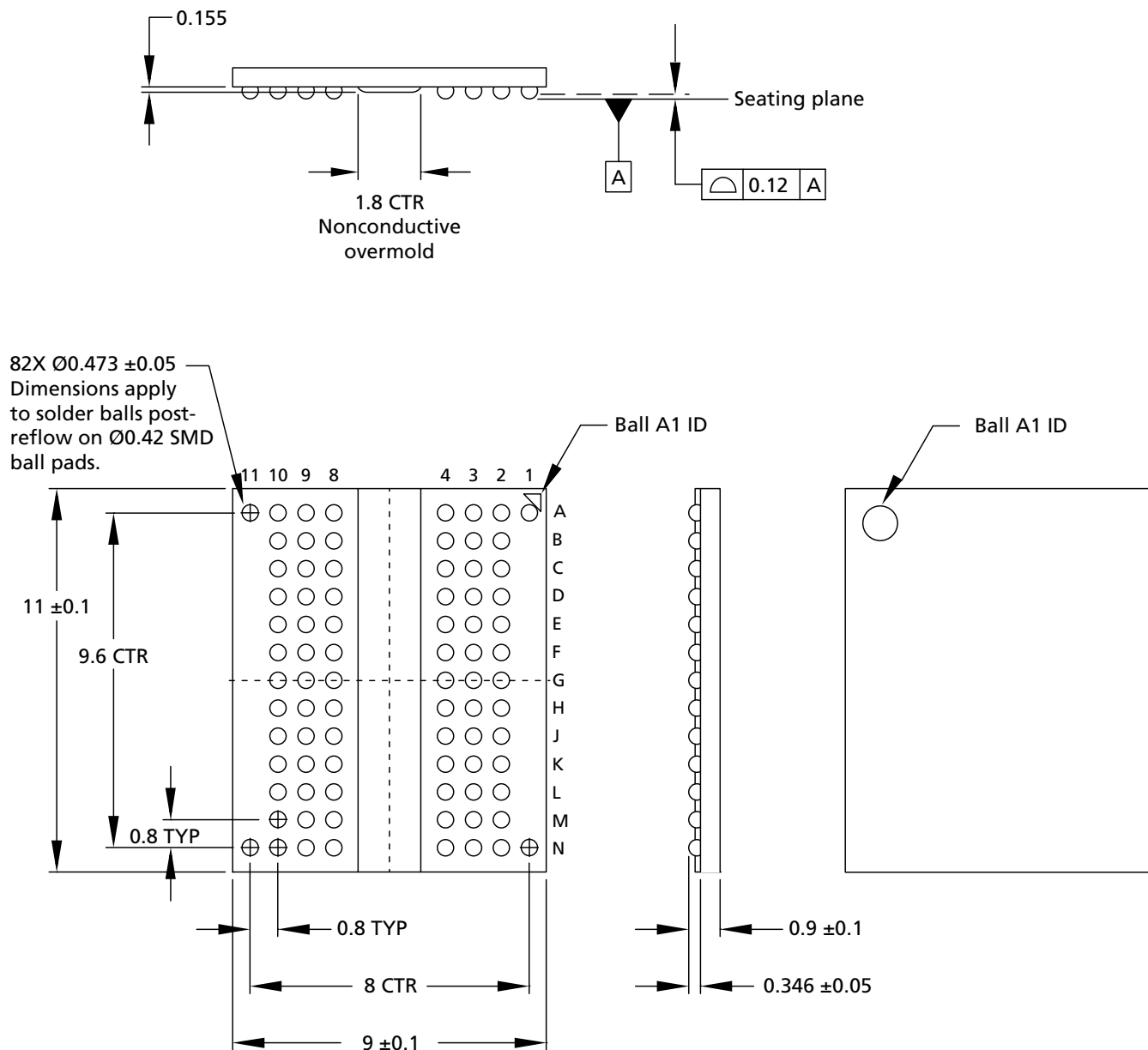
| Symbol                                       | Type         | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CK_t, CK_c                                   | Input        | <b>Clock:</b> CK_t and CK_c are differential clock inputs. All command/address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.                                                                                                                                                                                                                                                                                                                                                                                                 |
| CS_n                                         | Input        | <b>Chip Select:</b> All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code and is used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode, the CS_n input buffer operates with the same ODT and V <sub>REF</sub> parameters as configured by the CA_ODT strap setting or mode register. When in self refresh mode, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V <sub>DDQ</sub> . |
| DM_n, DMU_n, DML_n                           | Input        | <b>Input Data Mask:</b> DM_n is an input mask signal for write data. Input data is masked when DM_n is sampled LOW coincident with that input data during a write access. DM_n is sampled on both edges of DQS. DM_n is not supported on x4 devices. For x8 devices, the function of DM_n is enabled by the mode register. For x16 devices, the function of DMU_n/DML_n is enabled by the mode register.                                                                                                                                                                                     |
| CA[13:0]                                     | Input        | <b>Command/Address Inputs:</b> Command/Address (CA) signals provide the command and address inputs according to the Command Truth Table. Because some commands are multicyle, the pins may not be interchanged between devices on the same bus.                                                                                                                                                                                                                                                                                                                                              |
| RESET_n                                      | Input        | <b>Active Low Asynchronous Reset:</b> Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V <sub>DDQ</sub> .                                                                                                                                                                                                                                                                                                                                  |
| DQ                                           | Input/Output | <b>Data Input/Output:</b> Bidirectional data bus. If CRC is enabled via the mode register, CRC code is added at the end of a data burst.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DQS_t, DQS_c, DQSU_t, DQSU_c, DQSL_t, DQSL_c | Input/Output | <b>Data Strobe:</b> Output with read data, input with write data, edge-aligned with read data, centered in write data. For x16 devices, DQSL corresponds to the data on DQL0-DQL7; DQSU corresponds to the data on DQU0-DQU7. The data strobes DQS_t, DQSL_t and DQSU_t are paired with differential signals DQS_c, DQSL_c, and DQSU_c, respectively, to provide differential pair signaling to the system during reads and writes. The device supports differential data strobe only, not single-ended.                                                                                     |
| TDQS_t, TDQS_c                               | Output       | <b>Termination Data Strobe:</b> Applicable to x8 devices only. When enabled via the mode register, the device enables the same termination resistance function on TDQS_t/TDQS_c that is applied to DQS_t/DQS_c. When disabled via the mode register, DM/TDQS provides the data mask function depending on the MR setting; TDQS_c is not used. x4/x16 devices must disable the TDQS function via the mode register.                                                                                                                                                                           |
| ALERT_n                                      | Input/Output | <b>Alert:</b> If there is an error in CRC, ALERT_n drives LOW for the period time interval and returns HIGH. During the connectivity test mode, this pin functions as an input. Usage of this signal is system-dependent. In cases where this pin is not connected, ALERT_n must be bonded to V <sub>DDQ</sub> on the system board.                                                                                                                                                                                                                                                          |
| TEN                                          | Input        | <b>Connectivity Test Mode Enable:</b> A HIGH on this pin enables CONNECTIVITY TEST MODE operation along with other pins. It is a CMOS rail-to-rail signal with AC HIGH and LOW at 80% and 20% of V <sub>DDQ</sub> . Usage of this signal is system-dependent. This pin is pulled LOW internally with a weak pulldown resistor to V <sub>SS</sub> .                                                                                                                                                                                                                                           |

**Table 4: Pinout Description (Continued)**

| Symbol    | Type         | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIR       | Input        | <b>Mirror:</b> Used to inform the system that this device is being run in mirrored mode instead of standard mode. With the MIR pin connected (strapped) to $V_{DDQ}$ , the device internally swaps even-numbered CA with the next higher odd-number CA. The MIR pin must be tied to $V_{SS}$ if no CA mirror is required. Mirror pair examples: CA2 with CA3 (not CA1) CA4 with CA5 (not CA3). Note: the CA[13] function is only relevant for certain densities (including stacking). In the case that CA[13] is not used, its ball location, considering whether MIR is used or not, should be connected (strapped) to $V_{DDQ}$ . No active signaling requirements required. |
| CAI       | Input        | <b>Command and Address Inversion:</b> With this pin connected (strapped) to $V_{DDQ}$ , the device internally inverts the logic level present on all CA signals. The CAI pin must be connected to $V_{SS}$ if no CA inversion is required. No active signaling requirements required.                                                                                                                                                                                                                                                                                                                                                                                          |
| CA_ODT    | Input        | <b>ODT for Command and Address:</b> Apply Group A settings if the pin is connected (strapped) to $V_{SS}$ ; apply Group B settings if the pin is connected (strapped) to $V_{DDQ}$ . See the mode register defaults table for details. No active signaling requirements required.                                                                                                                                                                                                                                                                                                                                                                                              |
| LBDQ      | Output       | <b>Loopback Data Output:</b> The output of this device on the Loopback Output Select defined in MR53:OP[4:0]. When loopback is enabled, it is in driver mode using the default RON described in the Loopback Function section. When loopback is disabled, the pin is either terminated or High-Z based on MR36:OP[2:0].                                                                                                                                                                                                                                                                                                                                                        |
| LBDQS     | Output       | <b>Loopback Data Strobe Output:</b> A single-ended strobe with the rising edged aligned with loopback data edge, falling edge aligned with data center. When loopback is enabled, it is in driver mode using the default RON described in the Loopback function section. When loopback is disabled, the pin is either terminated or High-Z based on MR36:OP[2:0].                                                                                                                                                                                                                                                                                                              |
| RFU       | Input/Output | Reserved for future use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DNU       |              | Do not use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| NF        |              | <b>No function:</b> Internal connection is present but has no function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $V_{DDQ}$ | Supply       | DQ power supply; 1.1V nominal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $V_{DD}$  | Supply       | Power supply; 1.1V nominal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $V_{SS}$  | Supply       | Ground                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $V_{PP}$  | Supply       | Activating power supply; 1.8V nominal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ZQ        | Reference    | Reference pin for ZQ calibration. This ball is tied to an external 240 ohm resistor (RZQ), which is tied to $V_{SS}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

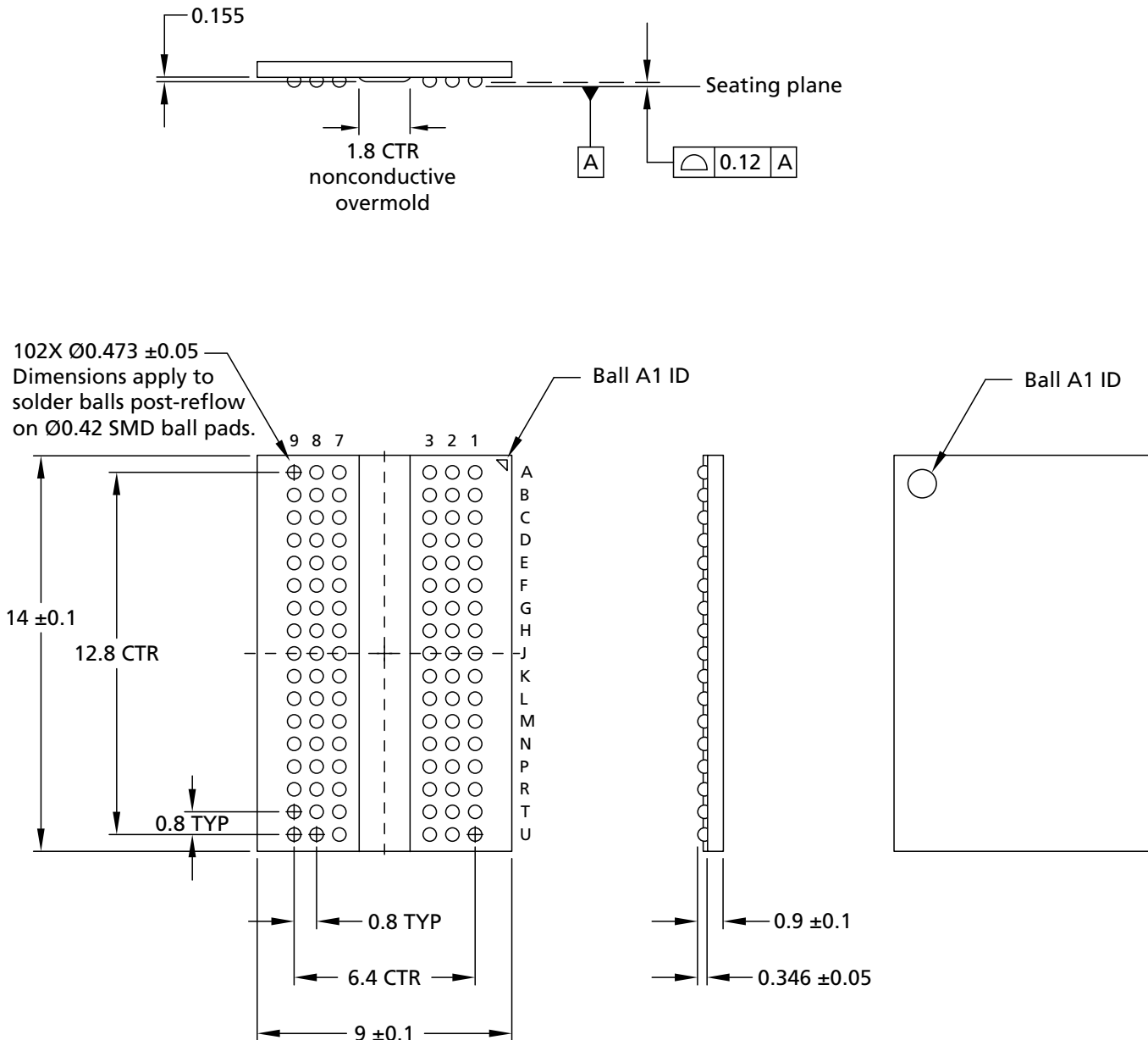
## Package Dimensions

**Figure 7: 82-Ball VFBGA – MO-210-AN (x4/x8)**



- Notes: 1. All dimensions are in millimeters.  
2. Solder ball material: SAC305 (96.5% Sn, 3% Ag, 0.5% Cu).

**Figure 8: 102-Ball VFBGA – MO-210-AT (x16)**



- Notes: 1. All dimensions are in millimeters.  
2. Solder ball material: SAC305 (96.5% Sn, 3% Ag, 0.5% Cu).

**Table 5: Package Thermal Resistance Characteristics**

| Die Revision | Package       | Parameter              | Value | Unit                 | Symbol        |
|--------------|---------------|------------------------|-------|----------------------|---------------|
| Rev A        | 82-ball "HB"  | Junction-to-case (TOP) | 2.6   | $^{\circ}\text{C/W}$ | $\theta_{JC}$ |
|              |               | Junction-to-board      | 12.8  | $^{\circ}\text{C/W}$ | $\theta_{JB}$ |
|              | 102-ball "HC" | Junction-to-case (TOP) | 2.6   | $^{\circ}\text{C/W}$ | $\theta_{JC}$ |
|              |               | Junction-to-board      | 12.8  | $^{\circ}\text{C/W}$ | $\theta_{JB}$ |

## DDR5 IDD,IPP,IDDQ Current Limits

DDR5 SDRAM current limits are measured and categorized based on the definitions found in the DDR5 Product Core data sheet. Refer to the IDD and IDDQ specification parameters and test conditions for details related to each current limit. Notes 1 and 2 apply to entire table

**Table 6: DDR5 IDD, IPP, IDDQ Current Limits – 16Gb Die Revision A**

| Parameter | Width | DDR5-4800 | Unit | Notes |
|-----------|-------|-----------|------|-------|
| IDD0      | x4    | 103       | mA   |       |
|           | x8    |           |      |       |
|           | x16   | 122       |      |       |
| IPP0      | x4    | 8         | mA   |       |
|           | x8    |           |      |       |
|           | x16   | 10        |      |       |
| IDDQ0     | x4    | 31        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD0F     | x4    | 142       | mA   |       |
|           | x8    |           |      |       |
|           | x16   | 164       |      |       |
| IPP0F     | x4    | 10        | mA   |       |
|           | x8    |           |      |       |
|           | x16   | 13        |      |       |
| IDDQ0F    | x4    | 33        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD2N     | x4    | 92        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IPP2N     | x4    | 6         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ2N    | x4    | 33        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD2NT    | x4    | 149       | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |



# 16Gb DDR5 SDRAM Die Rev A DDR5 IDD,IPP,IDDQ Current Limits

**Table 6: DDR5 IDD, IPP, IDDQ Current Limits – 16Gb Die Revision A**

| Parameter | Width | DDR5-4800 | Unit | Notes |
|-----------|-------|-----------|------|-------|
| IPP2NT    | x4    | 6         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ2NT   | x4    | 32        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD2P     | x4    | 88        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IPP2P     | x4    | 6         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ2P    | x4    | 24        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD3N     | x4    | 142       | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IPP3N     | x4    | 7         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ3N    | x4    | 31        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD3P     | x4    | 140       | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IPP3P     | x4    | 7         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ3P    | x4    | 24        | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDD4R     | x4    | 318       | mA   |       |
|           | x8    | 377       |      |       |
|           | x16   | 530       |      |       |



# 16Gb DDR5 SDRAM Die Rev A DDR5 IDD,IPP,IDDQ Current Limits

**Table 6: DDR5 IDD, IPP, IDDQ Current Limits – 16Gb Die Revision A**

| Parameter | Width | DDR5-4800 | Unit | Notes |
|-----------|-------|-----------|------|-------|
| IPP4R     | x4    | 9         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ4R    | x4    | 43        | mA   |       |
|           | x8    | 57        |      |       |
|           | x16   | 92        |      |       |
| IDD4RC    | x4    | 328       | mA   |       |
|           | x8    | 389       |      |       |
|           | x16   | 546       |      |       |
| IPP4RC    | x4    | 9         | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |
| IDDQ4RC   | x4    | 44        | mA   |       |
|           | x8    | 57        |      |       |
|           | x16   | 93        |      |       |
| IDD4W     | x4    | 345       | mA   |       |
|           | x8    | 349       |      |       |
|           | x16   | 479       |      |       |
| IPP4W     | x4    | 36        | mA   |       |
|           | x8    | 37        |      |       |
|           | x16   | 64        |      |       |
| IDDQ4W    | x4    | 116       | mA   |       |
|           | x8    | 198       |      |       |
|           | x16   | 348       |      |       |
| IDD4WC    | x4    | 311       | mA   |       |
|           | x8    | 316       |      |       |
|           | x16   | 421       |      |       |
| IPP4WC    | x4    | 35        | mA   |       |
|           | x8    | 37        |      |       |
|           | x16   | 64        |      |       |
| IDDQ4WC   | x4    | 116       | mA   |       |
|           | x8    | 194       |      |       |
|           | x16   | 344       |      |       |
| IDD5B     | x4    | 277       | mA   |       |
|           | x8    |           |      |       |
|           | x16   |           |      |       |



# 16Gb DDR5 SDRAM Die Rev A DDR5 IDD,IPP,IDDQ Current Limits

**Table 6: DDR5 IDD, IPP, IDDQ Current Limits – 16Gb Die Revision A**

| Parameter      | Width | DDR5-4800 | Unit | Notes |
|----------------|-------|-----------|------|-------|
| IPP5B          | x4    | 28        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDDQ5B         | x4    | 32        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDD5C          | x4    | 135       | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IPP5C          | x4    | 12        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDDQ5C         | x4    | 32        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDD5F          | x4    | 262       | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IPP5F          | x4    | 26        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDDQ5F         | x4    | 32        | mA   |       |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDD6N(0-85C)   | x4    | 102       | mA   | 3,4   |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IPP6N (0-85C)  | x4    | 15        | mA   | 3,4   |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDDQ6N (0-85C) | x4    | 16        | mA   | 3,4   |
|                | x8    |           |      |       |
|                | x16   |           |      |       |
| IDD6E (85-95C) | x4    | 200       | mA   | 4,5   |
|                | x8    |           |      |       |
|                | x16   |           |      |       |

**Table 6: DDR5 IDD, IPP, IDDQ Current Limits – 16Gb Die Revision A**

| Parameter       | Width | DDR5-4800 | Unit | Notes |
|-----------------|-------|-----------|------|-------|
| IPP6E (85-95C)  | x4    | 25        | mA   | 4,5   |
|                 | x8    |           |      |       |
|                 | x16   |           |      |       |
| IDDQ6E (85-95C) | x4    | 19        | mA   | 4,5   |
|                 | x8    |           |      |       |
|                 | x16   |           |      |       |
| IDD7            | x4    | 448       | mA   |       |
|                 | x8    | 502       |      |       |
|                 | x16   | 775       |      |       |
| IPP7            | x4    | 23        | mA   |       |
|                 | x8    | 23        |      |       |
|                 | x16   | 35        |      |       |
| IDDQ7           | x4    | 50        | mA   |       |
|                 | x8    | 64        |      |       |
|                 | x16   | 100       |      |       |
| IDD8            | x4    | 80        | mA   |       |
|                 | x8    |           |      |       |
|                 | x16   |           |      |       |
| IPP8            | x4    | 6         | mA   |       |
|                 | x8    |           |      |       |
|                 | x16   |           |      |       |
| IDDQ8           | x4    | 19        | mA   |       |
|                 | x8    |           |      |       |
|                 | x16   |           |      |       |

- Notes: 1. Some  $I_{DD}$  currents are higher for x16 organization due to larger page-size architecture.  
2. Maximum values for  $I_{DD}$  currents considering worst-case conditions of process, temperature, and voltage.  
3. Applicable for MR4:OP[2:0]=001b, 010b.  
4. Supplier data sheets include a maximum value for  $I_{DD6}$ .  
5. Applicable for MR4:OP[2:0]=011b, 100b, 101b.

## Revision History

### Rev. D – 02/2023

- Functional block diagrams updated

### Rev. C – 01/2023

- Removed 52B, 56B speed bins (not supported in 16Gb Die Revision A).
- Added functional block diagrams.
- Moved thermal characteristics table under package dimension topic.

### Rev. B – 10/2021

- Removed Micron Confidential marking.
- Updated DDR5 Function Matrix table to add default mode register states.

### Rev. A – 09/2021

- Initial release

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This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change, as further product development and data characterization sometimes occur.

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